

2(a)	the point where (all) the weight (of the object) is taken to act	B1
2(b)(i)	(54×0.45) or (2.4×0.95) or $(T \sin 30^\circ \times 1.3)$	C1
	$(54 \times 0.45) = (2.4 \times 0.95) + (T \sin 30^\circ \times 1.3)$	C1
	$T = 34 \text{ N}$	A1
2(b)(ii)	resultant moment = $(54 \times 0.45) - (2.4 \times 0.95)$ or $(34 \sin 30^\circ \times 1.3)$ = 22 N m	A1
2(c)(i)	$(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$	C1
	= 2.4×1.8	A1
	= 4.3 J	

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Question	Answer	Marks
2(c)(ii)	$E = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times (2.4 / 9.81) \times 3.4^2$ $= 1.4 \text{ J (at X)}$	C1
	kinetic energy at Y $= 4.3 + 1.4$ $= 5.7 \text{ J}$	A1
	or	
	$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mg(\Delta)h$	(C1)
	$v^2 = 3.4^2 + 2 \times 9.81 \times 1.8$ $v^2 = 46.9$ so $v = 6.85 \text{ (ms}^{-1}\text{)}$ $\text{KE} = \frac{1}{2} \times (2.4 / 9.81) \times 6.85^2$	(C1)
	$= 5.7 \text{ J}$	(A1)
2(c)(iii)	no variation or acceleration is (always) vertically downwards	B1
2(c)(iv)	horizontal straight line at a non-zero value of velocity	B1